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Subject Code/Name: MLA0305-Reinforcement Learning

EXPERIMENT - 7

Temporal Difference Learning Using TD(0) Algorithm

Aim

To implement the TD(0) Temporal Difference Learning algorithm and estimate the value of states through continuous interaction with the environment.

Procedure

1. Define the states, actions, and rewards.
2. Initialize the state-value function.
3. Start the agent from the initial state.
4. Select an action according to the policy.
5. Observe the reward and next state.
6. Update the value using the TD(0) equation.
7. Repeat the process for multiple episodes.
8. Visualize the learned state values.

Code:

```
# =====  
# EXPERIMENT 7: TEMPORAL DIFFERENCE LEARNING - TD(0)  
# =====  
  
import numpy as np  
import matplotlib.pyplot as plt  
  
np.random.seed(42)  
  
states = ["Start", "Middle", "Goal"]  
gamma = 0.9    # Discount factor  
alpha = 0.1    # Learning rate  
episodes = 100
```

```

# Initial state values
V = {s: 0.0 for s in states}

# Fixed policy
policy = {
    "Start": "Move",
    "Middle": "Move",
    "Goal": "Stay"
}

# Reward function
def reward(state):
    if state == "Start":
        return 1
    if state == "Middle":
        return 10
    return 0

# Transition function
def transition(state):
    if state == "Start":
        return "Middle"
    if state == "Middle":
        return "Goal"
    return "Goal"

# Store values for visualization
history = []

# TD(0) Learning
for episode in range(epochs):

    state = "Start"

    while state != "Goal":

```

```

next_state = transition(state)
r = reward(state)

# TD(0) Update:
#  $V(s) = V(s) + \alpha[R + \gamma V(s') - V(s)]$ 

V[state] += alpha * (
    r + gamma * V[next_state] - V[state]
)

state = next_state

history.append([V[s] for s in states])

# Display results
print("TD(0) State Values")
print("-" * 30)

for s in states:
    print(f"{s}: {V[s]:.3f}")

# Visualization
history = np.array(history)

plt.figure(figsize=(9, 5))

for i, s in enumerate(states):
    plt.plot(
        history[:, i],
        label=s
    )

plt.title("TD(0) Value Function Convergence")
plt.xlabel("Episode")

```

```
plt.ylabel("State Value")
plt.legend()
plt.grid(True)
plt.show()

# Final value visualization
plt.figure(figsize=(7, 5))

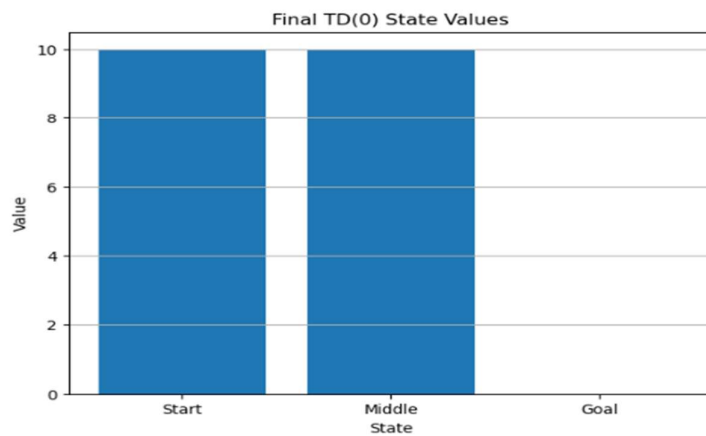
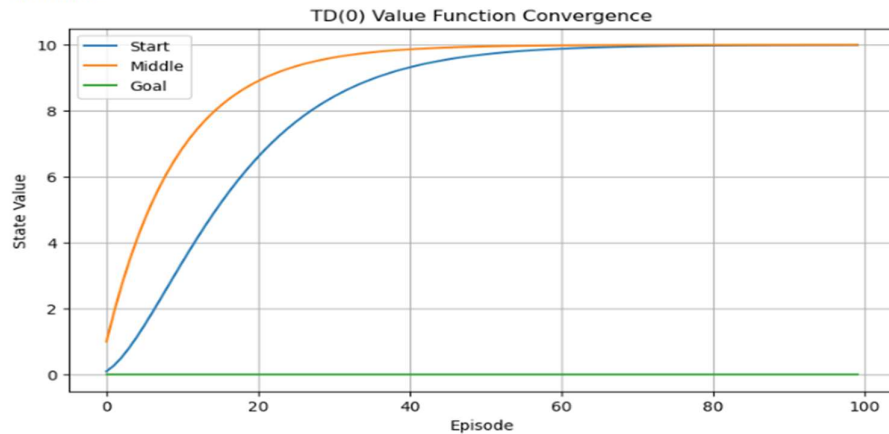
plt.bar(
    states,
    [V[s] for s in states]
)

plt.title("Final TD(0) State Values")
plt.xlabel("State")
plt.ylabel("Value")
plt.grid(axis="y")
plt.show()
```

OUTPUT:-

TD(0) State Values

Start: 9.997
Middle: 10.000
Goal: 0.000



Visualization

The program produces two graphs:

1. TD(0) Value Function Convergence – shows how the estimated state values change over successive episodes.
2. Final TD(0) State Values – shows the final estimated value of Start, Middle, and Goal states.

Summary

The TD(0) algorithm was implemented using incremental updates based on the current reward and estimated value of the next state. The state values gradually converge through repeated learning episodes.

Result

Thus, Temporal Difference Learning using the TD(0) algorithm was successfully implemented, and the convergence of the state-value function was visualized.